NETFLIX MOVIES & TV SHOWS

DATA ANALYSIS AND VISUALIZATION

A Python-Based Academic Project

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Abstract

This project demonstrates descriptive data analysis and visualization using a streaming-catalog dataset. It covers data inspection, cleaning, summary statistics, categorical frequency analysis, and charts using Python libraries. The bundled dataset is synthetic and educational; its values are not claims about Netflix's real catalog.

1. Introduction

Streaming platforms contain extensive catalogs of movies and television programs. Catalog metadata can be analyzed to understand content composition, release-year patterns, country representation, genres, and ratings. Exploratory data analysis helps convert raw records into understandable information.

2. Problem Statement

This project investigates a catalog dataset to answer five questions: What is the distribution of Movies and TV Shows? How are titles distributed by release year? Which country and genre categories are most frequent? How do ratings vary by content type? What data-quality issues require cleaning?

3. Objectives

The objectives are to load and inspect a CSV dataset, identify missing values and duplicates, clean selected fields, calculate descriptive summaries, create visualizations, interpret patterns, and document limitations.

4. Tools and Technologies

Python is the programming language. Pandas is used for data loading, transformation, and summaries. Matplotlib and Seaborn are used for charts. Jupyter Notebook combines executable code, results, and explanations.

5. Dataset Description

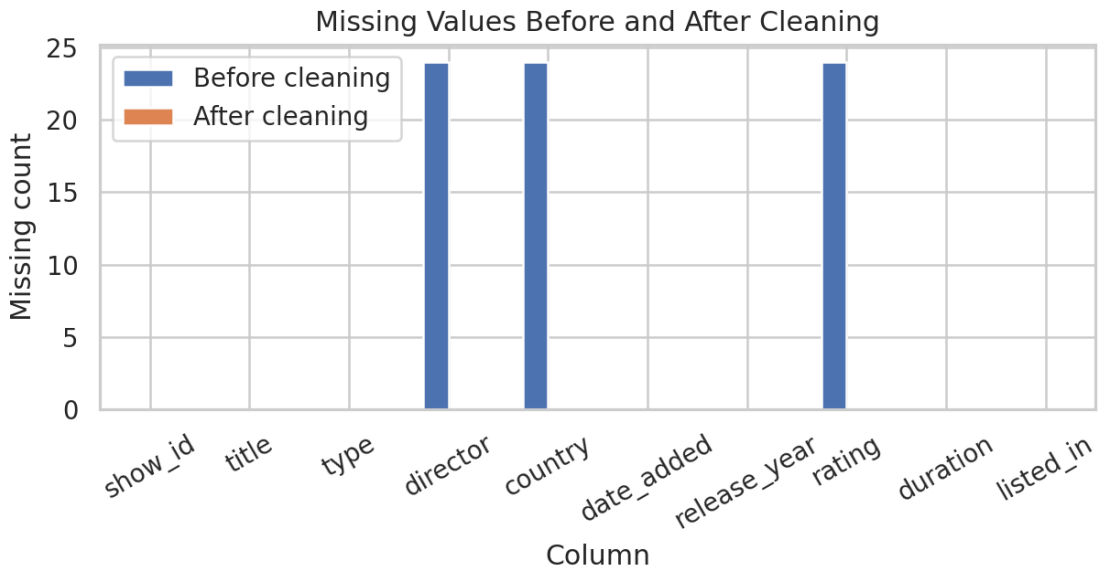
The bundled file contains 600 synthetic records with identifiers, titles, content type, director, country, date added, release year, rating, duration, and genre. It is included so the analysis is reproducible without external downloads. It is not the official Netflix dataset.

6. Methodology

The workflow includes importing libraries, reading the CSV, examining dimensions and data types, measuring missing values, removing duplicate rows, filling selected categorical gaps, converting year/date fields, calculating frequencies and descriptive statistics, plotting distributions, and interpreting results.

7. Data Cleaning

Duplicate records are removed. Missing director, country, and rating values are filled with 'Not Specified' to retain records while making missing information explicit. Release year is converted to numeric and date-added values are parsed as dates. Cleaning decisions should be revisited when using a real dataset.

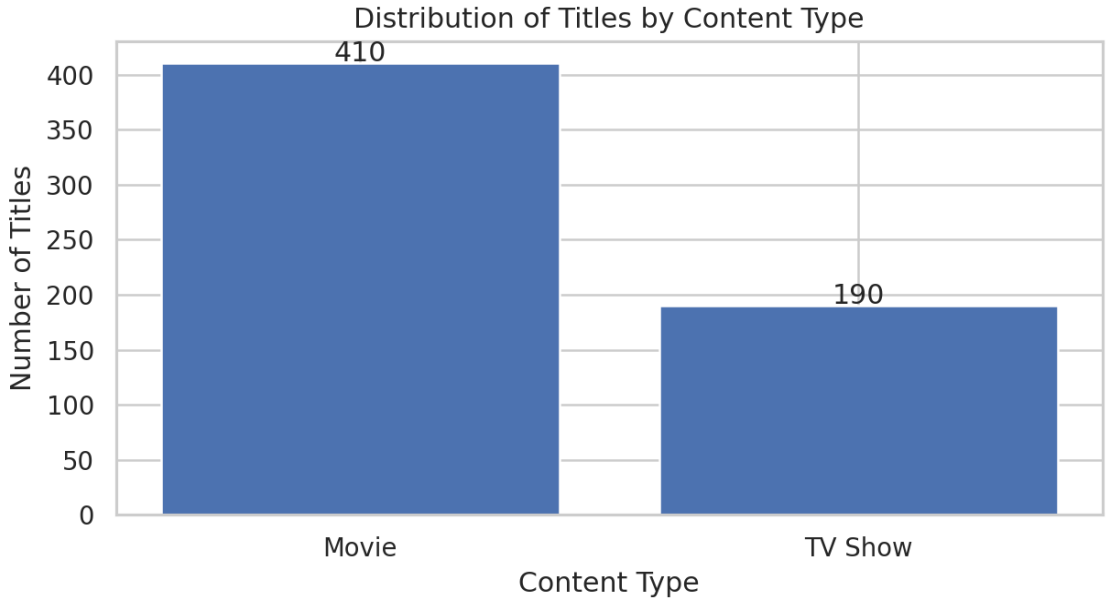


8. Descriptive Analysis

Frequency counts summarize Movies and TV Shows, countries, genres, and ratings. The release-year describe() output reports count, mean, standard deviation, minimum, quartiles, and maximum. These measures offer a concise numerical overview.

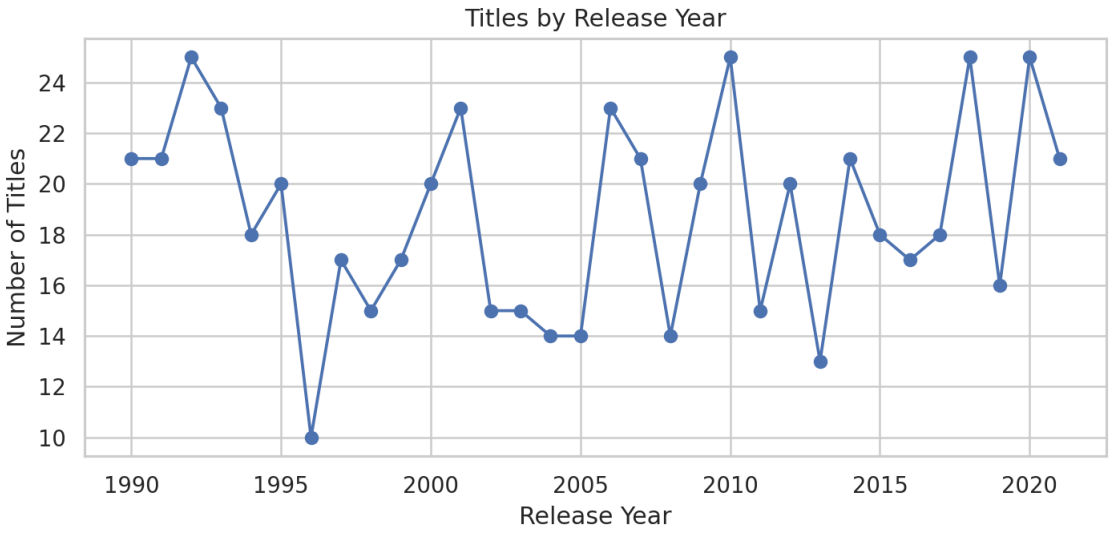
9. Content Type Visualization

The bar chart compares Movie and TV Show counts. It makes the relative frequency of two categorical groups easy to compare. The exact counts are computed from the included sample when the notebook is run.



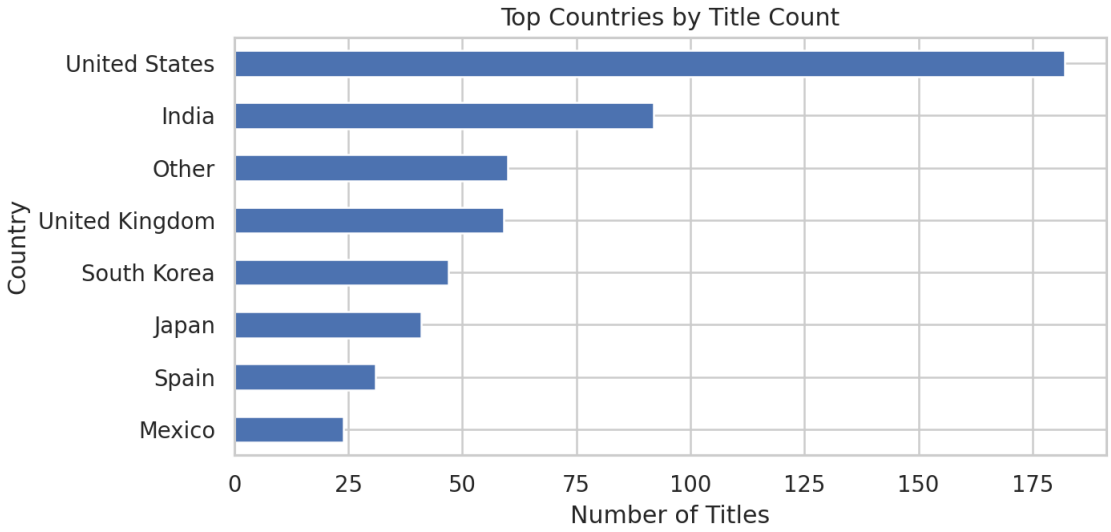
10. Release Year Visualization

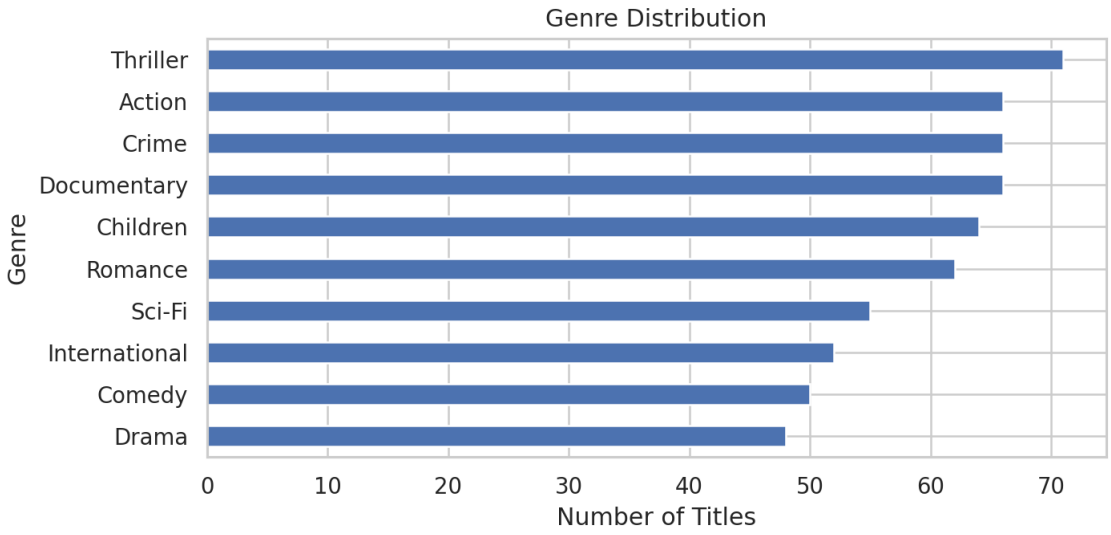
The line chart displays the number of titles for each release year. It helps identify the temporal distribution and years with relatively higher or lower representation. Because the data is synthetic, any apparent peaks are illustrative.



11. Country and Genre Visualization

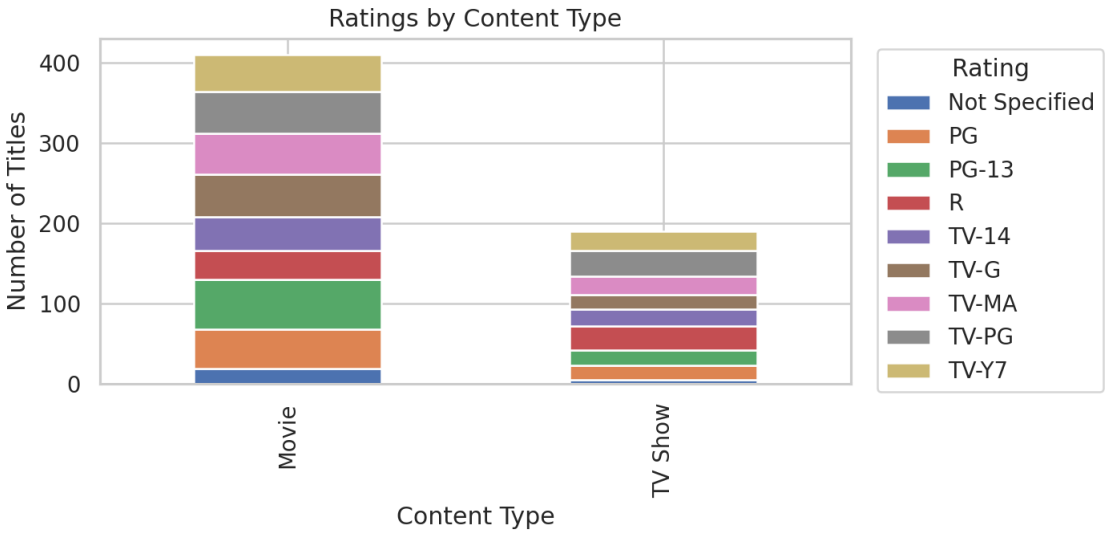
Horizontal bar charts compare country and genre frequencies. In authentic catalog data, country and genre fields can contain multiple labels per record; these should be split and normalized before calculating individual contributions.





12. Rating Visualization

The stacked bar chart compares rating categories across content types. It supports descriptive comparison of rating composition but does not imply that content type causes a particular rating.



13. Results and Key Findings

The notebook computes the exact content-type totals, release-year range and summary statistics, most frequent countries and genres, and rating-by-type table. These outputs demonstrate how descriptive analysis answers the project questions. As the dataset is synthetic, results must be labeled illustrative rather than real Netflix findings.

14. Limitations

The dataset is synthetic, so results cannot represent Netflix's actual catalog. Country and genre fields are modeled as single categories. The project is descriptive; it does not prove causal relationships or build a predictive model. A real-world study should cite its source and validate data quality.

15. Future Scope

The project can be extended with an authentic, cited dataset, normalization of multi-country and multi-genre values, interactive dashboard filters, analysis of date-added trends, and predictive modeling if a valid target variable and adequate data are available.

16. Conclusion

This project demonstrates a complete introductory data analysis pipeline, from loading and cleaning data to summary statistics, charts, and interpretation. The code is reusable and can be adapted to an authentic dataset by replacing the CSV and reviewing the cleaning process.

References

1. Python Documentation: https://docs.python.org/
2. Pandas Documentation: https://pandas.pydata.org/docs/
3. Matplotlib Documentation: https://matplotlib.org/stable/
4. Seaborn Documentation: https://seaborn.pydata.org/
5. Dataset: synthetic educational sample generated and included with this project.